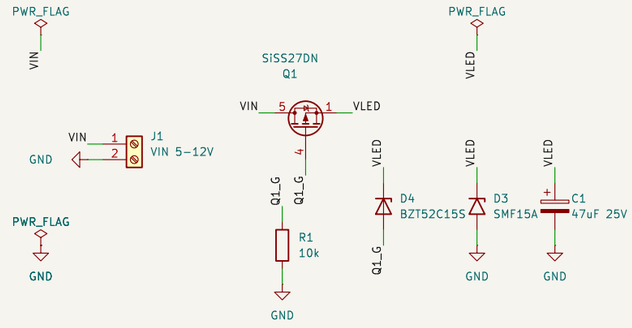
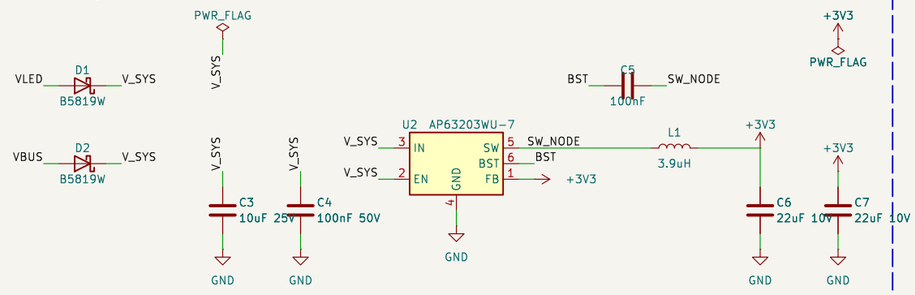
NGUON VAO 5–12V / CHONG CAM NGUOC / TVS

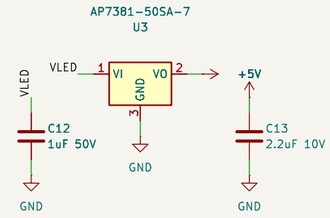


Q1 chong cam nguoc. Dong toi da 5 A. TVS D3 kep xung qua ap.
Khong co cau chi: dung nguon co bao ve ngan mach.

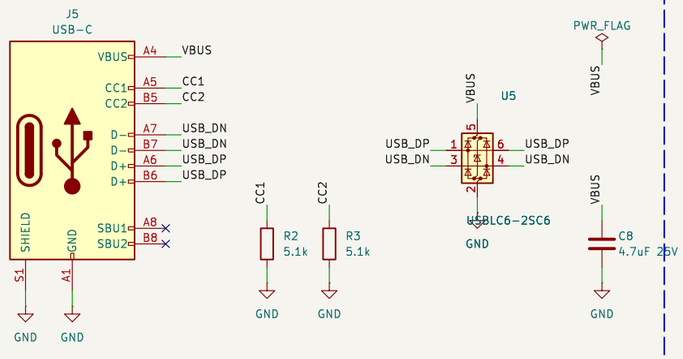
NGUON 3.3V (buck AP63203)



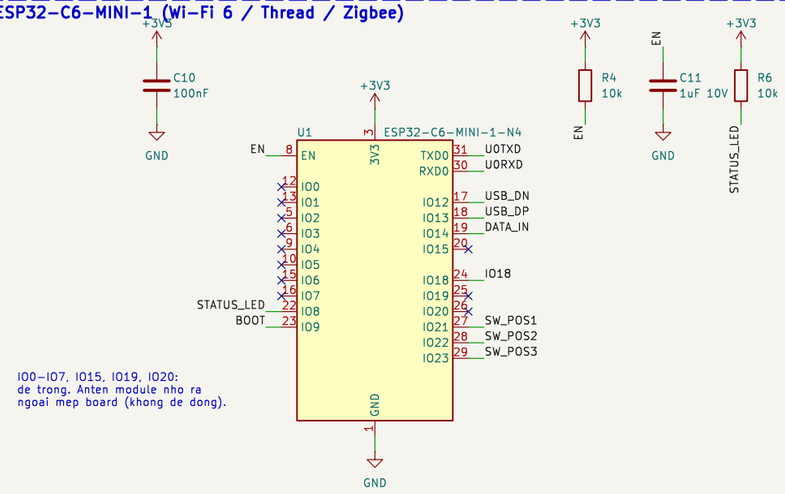
5V CHO IC DICH MUC (chi co khi co nguon LED)



USB-C: NAP CODE / DEBUG (USB Serial/JTAG)

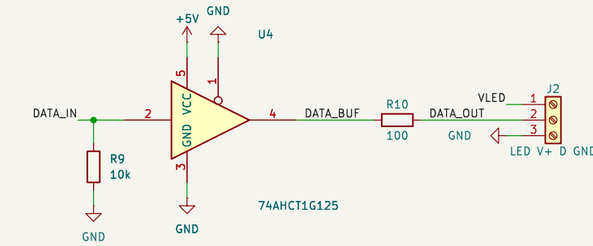


ESP32-C6-MINI-1 (Wi-Fi 6 / Thread / Zigbee)



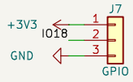
IO0–IO7, IO15, IO19, IO20:
de trong. Anten module nho ra
ngoai mep board (khong de dong).

DICH MUC 3.3V -> 5V + NGO RA WS2812



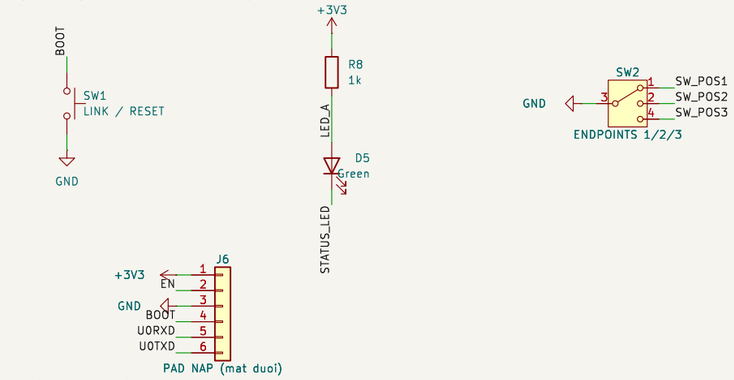
DATA 5V: WS2812B (5V) va WS2811 (12V) deu nhan muc 5V.

MO RONG: 1 GPIO TU DO (J7)



IO18: GPIO thuong, pull-up yeu chi trong luc reset.
Muc 3.3 V, khong chiu 5 V.

NUT / LED / CONG TAC 3 NAC / PAD NAP



SW1 (GPIO9 = BOOT): bam nhanh = bat/tat hoac vao che do ket noi; giu >5 s = factory reset.
Giu SW1 trong luc cap nguon = vao che do nap ROM.

SW2 = C&K PCM13SMTR, chan 3 la chan chung (noi GND). Nac 1 noi 1–3, nac 2 noi 2–3, nac 3 noi 3–4.
Firmware: IO21/IO22/IO23 la input, bat pull-up noi; chan nao doc 0 -> chia day LED thanh 1/2/3 doan,
moi doan la mot endpoint Matter. Khong chan nao doc 0 (dang gat giua 2 nac) -> giu gia tri cu.

Generated by tools/build_sch.py from tools/design.py
Input 5–12V, 1x WS2812/WS2811 output, 1 button, 3–position endpoint switch, 1 spare GPIO

Haxeko

Sheet: /		
File: c6_ws2812_1ch.kicad_sch		
Title: ESP32–C6 WS2812 1–channel Matter LED controller		
Size: A3	Date: 2026–11–09	**Rev: 3.0**
KiCad E.D.A. 9.0.9		Id: 1/1